ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
2	**(d)** 50	1/4 of 200 = 50 (Rryy).
3	**(c)** 1 : 1 : 1 : 1	Four gamete types x one.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Germ-cell DNA is unchanged, so acquired traits are not passed on.
5	**(d)** 1/4	Each birth is independent.
6	**(d)** new combinations of independently inherited genes	Independent assortment gives new combinations.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Masking explains absence of the recessive trait.
8	**(b)** Tt	1 : 1 ratio.
9	**(b)** environment affects expression of the trait	Phenotype depends on environment.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Equal gametes T and t give TT : Tt : tt = 1 : 2 : 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	TT fraction = 1/4 (1); 60 plants (1).	1+1
12	T is dominant over t (1); The tall allele masks the dwarf allele (1).	1+1
13	Ff x ff gives 1 : 1 (1); 2 children (1).	1+1
14	Ratio about 1 : 1 (1); Tall parent Tt, dwarf parent tt (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cross with green wrinkled (yyrr) (1); All yellow round offspring means YYRR (1); Other phenotypes in offspring mean the plant is heterozygous (1).	1+1+1
16	Sperm 23 (22 autosomes) (1); Egg 23 (22 autosomes); zygote 46 (44 autosomes) (1); Liver cell 46 (44 autosomes) (1).	1+1+1
17	A woman has two X chromosomes (XX) (1); Each egg receives only one of them (1); So each egg has one X and 22 autosomes (1).	1+1+1
	OR: Each pregnancy is independent (1); X and Y sperm are equally likely (1); Probability = 1/2 (1).	1+1+1
18	Green beetles are less easily seen (1); They survive and reproduce more (1); The proportion of green beetles increases (1).	1+1+1
	OR: Both parents are heterozygous (1); Each passes the recessive allele (1); The child is homozygous recessive (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	F1: TtRr, tall round (1); F2 ratio 9 : 3 : 3 : 1 (1); Tall round 900, tall wrinkled 300 (1); Dwarf round 300, dwarf wrinkled 100 (1); Dwarf round genotypes ttRR and ttRr (1).	1+1+1+1+1
	OR: First two plants: dwarf offspring means each has t (1); 1/4 dwarf shows both are Tt (1); Third plant crossed with tt gives all tall (1); So it must be TT (1); If it were Tt, half would be dwarf (1).	1+1+1+1+1
20	Tt x tt gives 1 tall : 1 dwarf (1); Rr x Rr gives 3 round : 1 wrinkled (1); Combine the independent ratios (1); Tall round 3, tall wrinkled 1, dwarf round 3, dwarf wrinkled 1 (1); Ratio 3 : 1 : 3 : 1 (1).	1+1+1+1+1
	OR: DNA copying errors (1); Random fusion of gametes (1); Independent assortment of genes (1); Variation helps in adapting (1); Example: people with different skin tones or blood groups (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1/4	Four equally likely types.
2	**(b)** Mother	Sons get the Y from the father.
3	**(d)** A is false but R is true.	A is false: Tt produces T and t gametes; R is true.
4	**(c)** 9	Nine distinct genotypes.
5	**(d)** 1/4	Both parents are Ff.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Independent assortment.
7	**(b)** 1280	1/16 of total = 80, total = 1280.
8	**(c)** it was ahead of the knowledge of chromosomes and genes	The genetic basis was unknown then.
9	**(d)** A is false but R is true.	A is false, sex is decided by the father; R is true.
10	**(d)** 600	Half of F2 are Tt.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New types: round green (3) and wrinkled yellow (3) (1); 6/16 = 3/8 (1).	1+1
12	Wrinkled yellow = 3/16 (1); 3/16 x 640 = 120 (1).	1+1
13	All offspring are black, so the black parent has only B (1); Black parent BB, brown parent bb (1).	1+1
14	Each gets one of the mother's two different X chromosomes (1); Autosomes also combine differently in each gamete (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parents are both Tt (1); Offspring TT, Tt, Tt, tt (1); Tall 3 : dwarf 1 (1).	1+1+1
16	Less hormone, so dwarf plant (1); Giving the hormone makes it taller (1); The offspring still inherit the dwarf allele (1).	1+1+1
17	Homozygous dominant: dominant trait (1); Heterozygous: dominant trait (1); Homozygous recessive: recessive trait (1).	1+1+1
	OR: Test parent is rryy (1); Offspring RrYy, Rryy, rrYy, rryy (1); Round yellow, round green, wrinkled yellow, wrinkled green in 1 : 1 : 1 : 1 (1).	1+1+1
18	He counted a large number of plants in each generation (1); Ratios such as 3 : 1 appeared (1); Ratios led to the idea of dominant and recessive factors (1).	1+1+1
	OR: At least one parent is homozygous dominant (1); The other can be TT or tt (1); Tall allele is dominant (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Attached is recessive (ff) (1); Both parents must be Ff (1); Children: FF, Ff, Ff, ff possible (1); Probability of free earlobes = 3/4 (1); Probability of attached = 1/4 (1).	1+1+1+1+1
	OR: Egg: 22 + X (1); Sperm: 22 + X or 22 + Y (1); Zygote: 44 + XX (girl) or 44 + XY (boy) (1); Half chromosomes from each parent (1); So child shows traits of both parents (1).	1+1+1+1+1
20	Each birth is an independent event (1); Probability of a boy is always 1/2 (1); Eggs have X only (1); Sperm may have X or Y (1); Father's sperm decides the sex, not the mother (1).	1+1+1+1+1
	OR: Ratio close to 1 : 1 : 1 : 1 (1); Parent RrYy forms four types of gametes (1); In equal proportion (1); Genes are inherited independently (1); Test parent rryy gives only ry (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 46 and 46	Both are diploid.
2	**(a)** Both A and R are true and R is the correct explanation of A.	True explanation.
3	**(d)** 1/16	(1/2)^4.
4	**(d)** there is only one parent and only copying errors	Single parent, little variation.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Gene combination explains it.
6	**(b)** Father	Y comes from the father.
7	**(b)** 1/16	ttrr.
8	**(a)** Both A and R are true and R is the correct explanation of A.	A single gamete type shows F1's gametes.
9	**(a)** Y chromosome	Sons are XY and get Y from the father.
10	**(a)** dwarf	Less hormone leads to a short plant.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	F1 is Tt (1); F2 tt probability is 1/4 (1).	1+1
12	Y sperm gives XY (1); Probability is 0 (the child is a boy) (1).	1+1
13	Each heterozygous gene gives 2 types (1); 2 x 2 x 2 = 8 types (1).	1+1
14	Recessive trait shows only when both alleles are recessive (1); So the genotype has to be homozygous recessive (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Muscle strength gained by exercise is an acquired trait (1); It does not change the DNA of the sperm (1); So it will not be inherited (1).	1+1+1
16	The ratio 3 : 1 is a probability (1); Small numbers may not show the ratio (1); It appears in a large number of offspring (1).	1+1+1
17	Nutrition affects growth (1); Good diet and health can increase height (1); Genes set the potential only (1).	1+1+1
	OR: Each child is a separate fertilisation (1); X and Y sperm are equally likely each time (1); Previous births do not affect the next (1).	1+1+1
18	Dwarf allele was only masked in Tt (1); F2 showed dwarf plants again (1); So it was not lost (1).	1+1+1
	OR: Child is a girl (1); Egg gives an X (1); Zygote is XX (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Round yellow = 9/16 (1); (b) RRYY, RRyy, rrYY, rryy (1); Each = 1/16 so 4/16 = 1/4 (1); (c) RrYy = 2/4 x 2/4 = 1/4 (1); Verify totals add to 16 (1).	1+1+1+1+1
	OR: Genotype is the same (1); Environment affects the expression (1); Differences in nutrition or sunlight (1); The phenotype therefore varies (1); Such differences are not inherited (1).	1+1+1+1+1
20	Genes for shape and colour are on different chromosomes (1); Alleles of each gene separate into gametes (1); Four gamete types in equal numbers (1); 16 random combinations (1); 9 : 3 : 3 : 1 ratio results (1).	1+1+1+1+1
	OR: Woman Rr, man rr (1); Gametes R, r and r (1); Children Rr and rr (1); Rr (positive) 1/2 (1); rr (negative) 1/2 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.